



Altima Hybrid



Sentra CA (USA)

The Environment

We have extended the Vision Zero concept to our work on environmental technologies. In the area of emissions, for example, we are investigating CO₂ and other substances with an environmental impact, either indirect or direct. Nissan's goal is simple: zero emissions. The primary focus for us has been CO₂ reduction, and we have been quite successful in this area.

Nissan is developing new hybrid technology as well. However, we do not believe this technology is sufficiently mature enough yet for wide application in the market. It would be easy to sell 1,000 or 10,000 cars, but that is neither an effective solution for the environment nor a financially viable proposition for a manufacturer.

Nissan's greatest strength is in current technologies such as the CVT, or continuously variable transmission. The CVT is a low-cost, advanced technology that can be applied to all types of vehicles to significantly and immediately reduce CO₂ emissions. In comparison to a hybrid electric vehicle, or HEV, a CVT-equipped car reduces CO₂ emissions by 20 percent. So if we sell five CVT-equipped vehicles, the effect would be the same as selling one hybrid car. Our current plan is to sell a million CVT vehicles, which would be equivalent to 200,000 HEVs—a significant figure.

We have to meet certain CO₂ emission levels. The first pillar in our efforts is to develop strong future technologies. To do this, Nissan must have a clear, precise vision of the future. We have come up with a number of specific scenarios for the next 40 or 50 years and considered the

technologies we will need to introduce to meet that vision. Over the next two decades we will significantly improve gasoline and diesel engines, which is where we can make an immediate impact. We are currently developing the next generation of gasoline HEVs. The generation after that will be diesel HEVs, which have even lower CO₂ emissions than gas-powered HEVs. The next step will be FCVs and pure electric vehicles. Nissan is actively working to further the diffusion of the fuel-cell stack we've developed in-house.

At the same time, we are taking a practical, proactive approach to the environment. We want to do good rather than just look good. The second pillar is to consider how we can actually introduce these new technologies to the world market. Advanced but expensive technology can't be applied on smaller, more economical cars, which are the cars most of us drive.

In the meantime, we have already had some notable successes in the environmental area. The Nissan Sentra CA, for example, was certified as the cleanest gasoline-powered car in the world, and our Bluebird Sylphy was recognized as the first-ever SU-LEV, or super-ultra-low emission vehicle. We are proud of these successes, but we are focusing on even more significant emissions reductions and fuel economy, continuing to develop advanced technologies that will bring us to our Vision Zero goal."

For more on environment at Nissan, please see the *2005 Nissan Sustainability Report*



X-TRAIL FCV



Nissan-original fuel cell stack



Continuously Variable Transmission (CVT)

CVT (Continuously Variable Transmission) enables a smooth, continuous transmission which not only enhances acceleration, but which also improves fuel economy for better environmental performance